

- Q.26 List two refractories used in blast furnace.
- Q.27 Explain soaking pits.
- Q.28 Explain sintering.
- Q.29 List two refractories used in cement plant.
- Q.30 Discuss insulating bricks.
- Q.31 Tell about ladle.
- Q.32 Discuss coke oven.
- Q.33 Explain hot metal mixer.
- Q.34 Write the uses of abrasives.
- Q.35 Explain composite material.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Describe the production, properties and uses of magnesia(magnesite) bricks.
- Q.37 Describe the production, properties and uses of cermet and composite material.
- Q.38 Discuss the refractories used in construction of blast furnace in glass industry in detail.

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Roll No.

5th Sem / Ceramic

Subject:- Ceramic Refractory Technology - II

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Which of the following is an example of specialrefractory?
 - a) Alumina b) Thoria
 - c) Fire clay d) Silica
- Q.2 Insulating refractories having
 - a) Low thermal conductivity
 - b) High thermal conductivity
 - c) Medium conductivity
 - d) none
- Q.3 $3\text{Al}_2\text{O}_3 \cdot 2\text{SiO}_2$ is
 - a) Alumina b) Ball clay
 - c) Mullite d) Dolomite
- Q.4 The largest consumer of refractories is the-
 - a) Cement plant b) Power plant
 - c) Metallurgical plant d) Fertiliser plant
- Q.5 Cermet are used in the
 - a) hearth of the blast furnace

- b) nuclear reactors, missiles & space crafts
 - c) insulation of high temperature furnaces
 - d) roof of electric furnaces
- Q.6 Carbon refractories are exclusively used in the
- a) Hearth of blast furnace
 - b) Side wall of soaking pits
 - c) Regenerators of coke oven
 - d) Walls of coke oven
- Q.7 High density refractory bricks have lower
- a) Spalling resistance
 - b) Slag penetration resistance
 - c) Fusion point
 - d) Thermal conductivity
- Q.8 Carbon has fusion point of
- a) 3600°C b) 600°C
 - c) 1600°C d) 1200°C
- Q.9 Refractory castable are used for
- a) Producing monolithic linings
 - b) Patch work
 - c) Both (a) and (b)
 - d) None
- Q.10 RUL stands for_____.
- a) Refractories under load
 - b) Refractoriness under load
 - c) Refractories upload
 - d) None

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SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Thoria is a _____ refractory.
- Q.12 Chromite refractoriness are _____ refractory.
- Q.13 Refractoriness of a typical silica brick corresponds to segar cone number, '32', which is equivalent to a temperature of _____°C.
- Q.14 Softening point of zirconia bricks is about _____°C.
- Q.15 Refractory bricks having lower porosity have good strength.(True/False)
- Q.16 Maximum alumina content in high alumina refractory can be as high as _____ percent.
- Q.17 CaO content in dolomite refractory can be _____ percent.
- Q.18 Porosity of insulating refractory brick should be high.(T/F)
- Q.19 TSR stands for_____.
- Q.20 SiC refractories are used in making of cutting wheels. (True/False)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 List the uses of mortar and cement.
- Q.22 List the properties of zirconia.
- Q.23 List the uses of crucible.
- Q.24 Discuss preparation of saggars.
- Q.25 Explain monolithic.

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